

Question

By utilizing cobalt metal ions to capture cyclic N_5^- ions in a solution containing N_5^- ions, a stable pure nitrogen anion metal salt complex **A** was successfully synthesized in air: $[\text{Co}(\text{N}_5)_x(\text{H}_2\text{O})_y] \cdot z\text{H}_2\text{O}$ (orange). It is known that \textcircled{1} the mass fractions are $\omega_{(\text{N})} = 40.82\%$ and $\omega_{(\text{O})} = 37.30\%$; \textcircled{2} if the coordinating groups in **A** are all regarded as spheres, then there are z C_2 axes perpendicular to the C_y axis in the inner sphere of **A**. Based on the above information, the correct option is:

- A.** In N_5^- , there exists a delocalized π bond Π_5^5 , which exhibits aromaticity.
- B.** $x + y + z = 8$
- C.** The molecular weight of **A** is 307.13
- D.** If all the ligand groups in **A** are regarded as spheres, then the inner sphere of **A** contains an antisymmetric axis I_8 .
- E.** If all ligand groups in **A** are considered as spheres, then all geometric isomers of the inner sphere of **A** are achiral.
- F.** If all the ligand groups in **A** are regarded as spheres, then the order of the point group to which the inner sphere of **A** belongs is 32.
- G.** If the ligand groups in **A** cannot be considered as spheres, then the highest-order rotation axis that can be contained in the inner sphere of **A** is C_4 .
- H.** If the coordinated groups in **A** cannot be considered as spheres, then at least 6 atoms in the inner sphere of **A** are coplanar.

Answer

Correct Answer: **E**

Detailed Explanation

Based on $\omega_{(N)} = 40.82\%$ and $\omega_{(O)} = 37.30\%$, it can be calculated that $n(N) : n(O) = 5 : 4$. Thus, it can be confirmed that $n(N_5^-) : n(H_2O) = 1 : 4$.

CHECKPOINT

1 PTS

$$n(N_5^-) : n(H_2O) = 1 : 4$$

First, set the compound as $Co(N_5)_x(H_2O)_{4x}$. When $x = 2$, the remaining molar mass is 58.98g/mol , which exactly corresponds to the atomic weight of one Co atom. Therefore, the chemical formula of **A** is $[Co(N_5)_2(H_2O)_y] \cdot (8 - y)H_2O$, with Co in the +2 oxidation state.

CHECKPOINT

2 PTS

The chemical formula of **A** is $[Co(N_5)_2(H_2O)_y] \cdot (8 - y)H_2O$, with Co in the +2 oxidation state

According to condition \textcircled{2}: If the coordinating groups in **A** are treated as spheres, the inner sphere of **A** contains z C_2 axes perpendicular to the C_y axis. y could be 2 or 4. However, when $y = 2$, it is impossible to have 6 C_2 axes perpendicular to the C_2 axis, so this case can be excluded. When $y = 4$ and $z = 4$, the trans-configuration of $Co(N_5)_2(H_2O)_4$ has 4 C_2 axes perpendicular to the C_4 axis, satisfying the condition. Therefore, the complex **A** is $\text{trans} - [Co(N_5)_2(H_2O)_4] \cdot 4H_2O$.

CHECKPOINT

1 PTS

$$y = 4, z = 4$$

CHECKPOINT

2 PTS

A is trans - $[\text{Co}(\text{N}_5)_2(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$

Now, evaluate each option one by one:

Option A: The N_5^- ion contains a delocalized π bond Π_5^6 and exhibits aromaticity. Each N atom contributes 1 electron, and the additional electron from the negative charge gives a total of 6 electrons, forming Π_5^6 . This satisfies Hückel's rule ($4 \times 2 + 2 = 6$), confirming aromaticity. Thus, the statement is incorrect.

CHECKPOINT

1 PTS

Hückel's rule $4 \times 2 + 2 = 6$, indicating aromaticity

Option B: $x + y + z = 8$. Based on the above derivation, $x + y + z = 10$, so the statement is incorrect.

CHECKPOINT

1 PTS

$$x + y + z = 10$$

Option C: The molecular weight of **A** is 307.13. According to the chemical formula, the calculated molecular weight is $M_A = 343.16$, so the statement is incorrect.

CHECKPOINT

1 PTS

Molecular weight $M_A = 343.16$

Option D: If the coordinating groups in **A** are treated as spheres, the inner sphere of **A** contains an improper rotation axis I_8 . If the coordinating groups are treated as spheres, the point group of the inner sphere is D_{4h} , which does not include an I_8 axis. Thus, the statement is incorrect.

CHECKPOINT

1 PTS

If the coordinating groups in **A** are treated as spheres, the inner sphere's point group is D_{4h} , which lacks an I_8 axis

Option E: If the coordinating groups in **A** are treated as spheres, all geometric isomers of the inner sphere of **A** are achiral. The cis-isomer has a C_{2v} point group, and the trans-isomer has a D_{4h} point group. Both contain mirror planes and are achiral, so the statement is correct.

CHECKPOINT

1 PTS

C_{2v} and D_{4h} point groups are both achiral

Option F: If the coordinating groups in **A** are treated as spheres, the order of the point group of the inner sphere is 32. If the coordinating groups are treated as spheres, the inner sphere's point group is D_{4h} , which has an order of 16. Thus, the statement is incorrect.

CHECKPOINT

1 PTS

D_{4h} has an order of 16

Option G: If the coordinating groups in **A** are not treated as spheres, the highest-order rotation axis in the inner sphere of **A** may be C_2 . Due to the structure of N_5^- , the C_4 axis is necessarily disrupted. If the water molecules are arranged with specific orientations, the original C_4 axis degenerates to C_2 , meaning the highest-order rotation axis possible is C_2 . Thus, the statement is incorrect.

CHECKPOINT

1 PTS

C_4 degenerates to C_2

Option H: If the coordinating groups in **A** are not treated as spheres, at least 7 atoms in the inner sphere of **A** are coplanar. One N_5^- group has 5 coplanar atoms, plus the central Co atom and one coordinating N atom from the other N_5^- group, totaling 7 coplanar atoms (1 Co and 6 N atoms). Thus, the statement is incorrect.

CHECKPOINT

1 PTS

1 Co atom and 6 N atoms, totaling 7 coplanar atoms

In conclusion, only option E is correct.